Teaching Statement

Stanisław Szufa

Teaching Philosophy

In my research, I prove properties of voting rules, and I also run them on real and synthetic data. I would like my students to do both.

- *Concepts come with data.* When I introduce a definition, students also apply it to a concrete instance. For example, students can load a real participatory budget from Pabulib, compute the outcomes of several voting rules, and only then study the axioms that explain why the outcomes differ.
- *Open-ended problems.* Alongside standard exercises, I will set problems without a known answer, such as finding the statistical model of votes that best describes a city's ballots. Students then have to choose a method themselves and argue for their conclusions.
- *Experimental code as part of the course.* Research-oriented courses rarely teach how to write clean, reproducible code for experiments. I will teach it using `prefsampling` and `mapof`, two Python libraries I co-developed.

Teaching and Mentoring Experience

At Jagiellonian University (2018–2021), I taught *Introduction to Programming* (a full-year course), *Logic and Set Theory*, and *Linear Algebra* (one semester each) to groups of about fifteen undergraduate students. For my research community, I co-authored the *Guide to Numerical Experiments on Elections* [C15] and co-developed `prefsampling` and `mapof`, and I co-presented the tutorial *Experiments in Social Choice* at IJCAI 2023, AAMAS 2023, and AAAI 2024. I have also given more than thirty research talks, which has made me comfortable explaining technical material to mixed audiences.

I have co-supervised more than ten BSc and MSc students, and work with students led to conference papers at AAAI 2025 [C19] and ECAI 2025 [C22]. I start each student with a well-defined question for which data and tools are at hand, so that a first result comes quickly, and then move to an open problem whose answer I do not know. I review experimental design, code, and writing as carefully as proofs. As President of the PhD Students' Association at Jagiellonian University, I led an elected body representing doctoral students.

Teaching at EPFL

I am prepared to teach core bachelor courses (introductory programming, discrete mathematics, algorithms) and master courses in artificial intelligence and algorithmic game theory. In large bachelor courses, I would use weekly programming and problem-solving exercises with quick feedback, automated where possible, and short problems during lectures that show me which ideas students have not yet understood. Exams would test both proofs and computation. I would like to develop two new courses:

- *Algorithms for Collective Decisions* (master): voting, participatory budgeting, fair division, and matching, with real data throughout and a final project on a real participatory budget.
- *AI and Democracy* (master seminar): AI-mediated deliberation, AI-written summaries of public consultations, and AI agents that vote on people's behalf, open to students from computer science and the social sciences.

Student projects could also be done with Swiss municipalities that run participatory budgets.